Case Study 1: Enterprise RAG Application for Secure Document Intelligence

◈ The Overview

I developed a **multi-tenant Retrieval-Augmented Generation (RAG) application** to help enterprise clients securely access, search, and generate insights from their internal documents such as RFPs, proposals, policies, and project records.
The solution enables users to ask natural-language questions and receive accurate, context-aware answers grounded strictly in their own data.

◈ The Challenges

- Client documents were **scattered across multiple formats** (PDFs, Word, internal files)
- No centralized **knowledge retrieval system**
- High risk of **data leakage** in generic AI tools
- Need for **tenant-wise isolation** and access control
- Prompt inconsistency across users and teams

◈ The Solution

We designed and implemented a **secure RAG architecture** using vector-based document indexing and Azure-hosted AI models.
The system retrieves the most relevant document chunks and feeds them into the AI model to ensure responses are accurate, traceable, and compliant with enterprise data policies.

◈ Solution Highlights

- Multi-tenant architecture with strict data isolation
- Vector search–based document retrieval
- Centralized **prompt management system** (read / write / restricted access)
- Support for structured outputs (summaries, reports, case studies)
- Environment-variable–driven deployment for scalability

◈ The Results

- Reduced document search time by **70%+**
- Improved response accuracy and consistency
- Enabled teams to generate compliant, client-ready content in minutes
- Increased confidence in AI usage due to secure data handling

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📌 Case Study 2: RFP & Proposal Intelligence Using RAG

◈ The Overview

To support proposal and bid teams, we implemented a RAG-based AI assistant that analyzes RFPs and generates structured summaries, compliance checklists, and staffing insights — all grounded in uploaded RFP documents.

◈ The Challenges

- Manual RFP review was time-consuming and error-prone
- Formatting and compliance requirements were often missed
- Proposal teams needed **repeatable, standardized outputs**
- Sensitive bid data could not be shared with public AI tools

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◈ The Solution

We built a **document-grounded AI workflow** that retrieves exact RFP clauses and instructions before generating summaries.
The system ensures that every AI response is directly tied to the source document.

◈ Solution Highlights

- Clause-level document retrieval
- Structured outputs for proposal managers
- Zero hallucination approach (answers only from documents)

- Word / PDF–ready content generation
- Tenant-specific document libraries

◈ The Results

- RFP analysis time reduced from days to **hours**
- Improved proposal compliance and accuracy
- Faster onboarding of new proposal team members
- Increased bid success confidence

📌 Case Study 3: Prompt Management & Governance in a RAG App

◈ The Overview

We implemented a **Prompt Governance framework** inside the RAG app to manage how AI behaves across different teams, roles, and tenants.

◈ The Challenges

- Users were creating inconsistent prompts
- No version control or approval mechanism
- Risk of prompts exposing sensitive logic
- Difficulty deploying prompts across environments

◈ The Solution

A centralized **prompt management module** was designed with role-based access, environment variables, and deployment-ready configurations.

◈ Solution Highlights

- Read-only vs editable prompt access

- Environment-based prompt deployment
- Reusable prompt templates
- Governance-ready design for enterprise clients

◈ The Results

- Consistent AI behavior across tenants
- Reduced operational risk
- Faster rollout of new AI capabilities
- Improved trust in AI-generated outputs